

Security & Privacy of Machine Learning (CS-452)

Project Proposal



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1 EXECUTIVE SUMMARY

This project proposes significant extensions to **Chimera**, a state-of-the-art log-based fault diagnosis framework introduced in the paper "*United We Stand: Towards End-to-End Log-based Fault Diagnosis via Interactive Multi-Task Learning*". While Chimera successfully combines anomaly detection and root cause localization through interactive multi-task learning, this project will extend its capabilities in three novel directions:

1. **Cross-Domain Generalization:** Adapting Chimera to work across diverse software system types (microservices, cloud platforms, IoT systems)
2. **Multi-Task Expansion:** Incorporating additional diagnostic tasks including log parsing, failure type classification, impact assessment, and remediation suggestion
3. **Advanced Task Interaction:** Designing novel interaction mechanisms including graph-based, hierarchical, and dynamic task prioritization

The proposed **Chimera-X** framework aims to achieve **15-25% improvement** over the original Chimera, establishing a new state-of-the-art in comprehensive log-based system reliability engineering.

2 BACKGROUND AND LITERATURE REVIEW

2.1 LOG-BASED FAULT DIAGNOSIS

Software systems generate massive volumes of logs that capture runtime behaviors. Analyzing these logs is crucial for:

- **Anomaly Detection (AD):** Identifying when system behavior deviates from normal patterns
- **Root Cause Localization (RCL):** Pinpointing the specific logs/events that caused failures

Current approaches treat these as separate tasks, leading to diagnostic bias accumulation and suboptimal performance.

2.2 THE CHIMERA FRAMEWORK

The original Chimera paper introduced several innovations:

Component	Description	Limitation
Interactive Multi-Task Learning	AD and RCL share knowledge bidirectionally	Limited to only 2 tasks
Shared-Private Encoder (SPE)	Disentangles task-specific and shared features	Tested only on 3 datasets
Cross-granularity Alignment	Aligns diagnostic results via JS divergence	Static interaction only

Sequence-driven Localizer	Trains without root cause labels	No dynamic task prioritization
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2.3 RESEARCH GAP

Despite Chimera's success, significant gaps remain:

Gap	Description	Why Important
Limited Domain Coverage	Tested only on BGL, Thunderbird, System A	Real-world systems are diverse
Task Incompleteness	Only 2 diagnostic tasks	SREs need comprehensive diagnosis
Static Interaction	Fixed interaction mechanism	System context changes dynamically
No Cross-Domain Adaptation	Assumes training/test from same domain	New systems lack labeled data

2.4 RELATED RECENT WORK

Paper	Approach	Limitation
LogLM (2024)	LLM-based log analysis	Computationally expensive
MIRTL (2025)	Multi-task representation learning	No task interaction mechanism
RIT (2024)	Relational interaction transformer	Fixed task relations
Eadro (2023)	End-to-end troubleshooting	Multi-source data required

3 PROBLEM STATEMENT

3.1 CORE PROBLEM

Current log-based fault diagnosis systems fail to provide **comprehensive, adaptable, and efficient** diagnosis due to:

- Diagnostic Bias Accumulation:** Errors from anomaly detection propagate to root cause localization
- Domain Specificity:** Models trained on one system fail on others (up to 17% performance drop)

3. **Task Isolation:** Related diagnostic tasks (parsing, classification, remediation) are treated separately
4. **Static Architectures:** Task interaction mechanisms don't adapt to changing system conditions

3.2 RESEARCH QUESTIONS

This project addresses the following research questions:

RQ	Question	Novelty Contribution
RQ1	How can Chimera be extended to work across diverse software system types without retraining?	Cross-domain adaptation
RQ2	Can incorporating additional diagnostic tasks improve overall fault diagnosis performance?	Multi-task expansion
RQ3	What interaction mechanisms enable optimal knowledge transfer among multiple diagnostic tasks?	Advanced interaction design
RQ4	How can task priorities be dynamically adjusted based on system state?	Dynamic task weighting

4 PROPOSED NOVELTY AND CONTRIBUTIONS

4.1 NOVELTY 1: CROSS-DOMAIN ADAPTATION MODULE

Problem: Chimera assumes training and testing data come from the same system distribution. Real-world deployment requires models that work on new, unseen systems.

Solution: A domain adaptation layer that learns system-invariant representations.

class CrossDomainChimera:

- Domain-adversarial training to remove system-specific features
- Few-shot adaptation for new systems with minimal labeled data
- Zero-shot inference capability

4.2 NOVELTY 2: MULTI-TASK EXPANSION

Problem: Real SRE workflows require more than just anomaly detection and root cause localization.

Solution: Extend Chimera to handle 4 additional tasks simultaneously:

New Task	Description	Output Format	Related Work
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Log Parsing	Extract structured templates	Event templates	Drain, Spell
Failure Classification	Categorize failure types	10+ failure categories	LogClass
Impact Assessment	Estimate severity and scope	Severity score (0-1) + affected components	ImpactRank
Remediation Suggestion	Recommend recovery actions	Action sequence	RemediateRL

4.3 NOVELTY 3: ADVANCED TASK INTERACTION MECHANISMS

Problem: Chimera uses a single, static interaction mechanism. Different system states require different interaction patterns.

Solution: Three novel interaction mechanisms:

A. Graph-Based Task Interaction

- Model tasks as nodes in a graph
- Learn task relations dynamically during training
- Message passing between related tasks

B. Hierarchical Task Grouping

- Group related tasks (detection tasks, localization tasks, analysis tasks)
- Intra-group dense interaction, inter-group sparse interaction
- Inspired by cognitive architectures

C. Dynamic Task Prioritization

- System state encoder monitors current logs
- Predicts which tasks are most important now
- Allocates computational resources accordingly

4.4 NOVELTY 4: UNIFIED EVALUATION BENCHMARK

Problem: No comprehensive benchmark exists for multi-task log diagnosis across domains.

Solution: Create and release:

Component	Details
Cross-Domain Dataset Suite	5 system types $\times$ 100K logs each
Multi-Task Labels	All 6 tasks labeled for 50K logs

Evaluation Protocol	Standardized metrics and splits
Baseline Implementations	All comparison methods open-sourced

5 METHODOLOGY

Phase 1: Foundation

- Set up Chimera codebase and reproduce results
- Collect and preprocess 5 new datasets (microservices, e-commerce, cloud, IoT, HPC)
- Implement unified data loader interface

Phase 2: Cross-Domain Adaptation

- Implement domain-adversarial training
- Design few-shot adaptation protocol
- Evaluate zero-shot performance across domains

Phase 3: Multi-Task Expansion

- Design and implement 4 additional task heads
- Create labeling pipeline for new tasks
- Implement task relation learning

Phase 4: Advanced Interaction

- Implement graph-based interaction
- Implement hierarchical grouping
- Implement dynamic prioritization
- Ablation studies and comparison

Phase 5: Evaluation and Documentation

- Comprehensive experiments
- Paper writing
- Code release and documentation

5.1 DATASETS

Dataset	System Type	Size
BGL	HPC Supercomputer	4M logs
Thunderbird	HPC Supercomputer	10M logs

System A	Cloud Platform	2M logs
GAIA	Microservices	1.5M logs
OpenStack	Cloud Infrastructure	3M logs
IoT-23	IoT Systems	2M logs

5.2 EVALUATION METRICS

Task	Metrics
Anomaly Detection	Precision, Recall, F1-Score
Root Cause Localization	HR@k, MAP@k, MRR (k=1,3,5)
Log Parsing	Accuracy, F1 (template-level)
Failure Classification	Accuracy, Macro-F1
Impact Assessment	MSE, Spearman Correlation
Remediation Suggestion	BLEU Score, Execution Success Rate
Cross-Domain	Zero-shot F1, Few-shot F1 (shots=10,50,100)
Efficiency	Inference time, Parameter count, FLOPs

6 CONCLUSION

This project proposes **Chimera-X**, a significant extension to the state-of-the-art Chimera framework for log-based fault diagnosis. By addressing three critical limitations—limited domain coverage, incomplete task set, and static interaction mechanisms—this project will deliver:

1. A **cross-domain adaptation module** enabling zero-shot deployment on new systems
2. A **multi-task expansion** incorporating 4 additional diagnostic tasks
3. **Three novel interaction mechanisms** for optimal knowledge transfer
4. A **comprehensive benchmark** for multi-task log diagnosis

The proposed extensions are expected to achieve **15-25% improvement** over the original Chimera, establishing a new state-of-the-art in comprehensive system reliability engineering. All code, datasets, and models will be open-sourced to enable reproduction and future research.

7 REFERENCES

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